

Step 1 – Create Database, Tables and Insert Class Data

```
1 • create database sweek3
2 ✖ use sweek3
3
4 ○ CREATE TABLE class (
5     id INT,
6     name VARCHAR(30)
7 );
8 ○ CREATE TABLE class_info (
9     id INT,
10    address VARCHAR(30)
11 );
12
13 • INSERT INTO class VALUES
14 (1, 'abhi'),
15 (2, 'adam'),
16 (4, 'alex');
```

Output



Action Output

	#	Time	Action	Message
✓	1	21:25:54	create database sweek3	1 row(s) affected
✓	2	21:26:08	use sweek3	0 row(s) affected
✓	3	21:26:22	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓	4	21:26:41	CREATE TABLE class_info (id INT, address VARCHA...	0 row(s) affected
✓	5	21:27:03	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'alex')	3 row(s) affected Records: 3 Dup

Step 2 – CROSS JOIN

```
16      (4, 'alex');
17 •   INSERT INTO class_info VALUES
18      (1, 'DELHI'),
19      (2, 'MUMBAI'),
20      (3, 'CHENNAI');
21
22 •   SELECT *
23      FROM class
24      CROSS JOIN class_info;
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	id	name	id	address
▶	4	alex	1	DELHI
	2	adam	1	DELHI
	1	abhi	1	DELHI
	4	alex	2	MUMBAI
	2	adam	2	MUMBAI
	1	abhi	2	MUMBAI
	4	alex	3	CHENNAI
	2	adam	3	CHENNAI
	1	abhi	3	CHENNAI

Result 1 x

Output

Action Output

	#	Time	Action	Message
✓	2	21:26:08	use week3	0 row(s) affected
✓	3	21:26:22	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓	4	21:26:41	CREATE TABLE class_info (id INT, address VARC...	0 row(s) affected
✓	5	21:27:03	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'al...	3 row(s) affected Reco
✓	6	21:27:33	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUM...	3 row(s) affected Reco
✓	7	21:27:45	SELECT * FROM class CROSS JOIN class_info LIMIT 0,...	9 row(s) returned

Step 3 - INNER JOIN

```
26 • SELECT *
27 FROM class
28 INNER JOIN class_info
29 ON class.id = class_info.id;
```

Result Grid				
Filter Rows:				
Export:  Wrap Cell Content: 				
	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI

Result 2 x

Output

 Action Output

	#	Time	Action	Message
✓	3	21:26:22	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓	4	21:26:41	CREATE TABLE class_info (id INT, address VARC...	0 row(s) affected
✓	5	21:27:03	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'al...	3 row(s) affected Reco
✓	6	21:27:33	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUM...	3 row(s) affected Reco
✓	7	21:27:45	SELECT * FROM class CROSS JOIN class_info LIMIT 0,...	9 row(s) returned
✓	8	21:28:18	SELECT * FROM class INNER JOIN class_info ON class...	2 row(s) returned

Step 4 – INNER JOIN with Selected Columns

```
31 • SELECT class.name,  
32         class_info.address  
33 FROM class  
34 INNER JOIN class_info  
35 ON class.id = class_info.id;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	name	address
▶	abhi	DELHI
	adam	MUMBAI

Result 3 x



Output



Action Output



	#	Time	Action	Message
✓	4	21:26:41	CREATE TABLE class_info (id INT, address VARC...	0 row(s) affected
✓	5	21:27:03	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'al...	3 row(s) affected Recor
✓	6	21:27:33	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUM...	3 row(s) affected Recor
✓	7	21:27:45	SELECT * FROM class CROSS JOIN class_info LIMIT 0,...	9 row(s) returned
✓	8	21:28:18	SELECT * FROM class INNER JOIN class_info ON class...	2 row(s) returned
✓	9	21:28:51	SELECT class.name, class_info.address FROM clas...	2 row(s) returned

Step 5 – NATURAL JOIN

36

37 • **SELECT** *

38 **FROM** class

39 **NATURAL JOIN** class_info;

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name	address
▶	1	abhi	DELHI
	2	adam	MUMBAI

Result 4 x



Rea

Output



Action Output



	#	Time	Action	Message
✓	5	21:27:03	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'al...	3 row(s) affected Records: 3
✓	6	21:27:33	INSERT INTO class_info VALUES (1, 'DELHI'), (2, 'MUM...	3 row(s) affected Records: 3
✓	7	21:27:45	SELECT * FROM class CROSS JOIN class_info LIMIT 0,...	9 row(s) returned
✓	8	21:28:18	SELECT * FROM class INNER JOIN class_info ON class...	2 row(s) returned
✓	9	21:28:51	SELECT class.name, class_info.address FROM clas...	2 row(s) returned
✓	10	21:29:21	SELECT * FROM class NATURAL JOIN class_info LIMIT...	2 row(s) returned

Step 6 - LEFT OUTER JOIN

```
40
47 • SELECT *
48 FROM class
49 LEFT OUTER JOIN class_info
50 ON class.id = class_info.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	4	alex	NULL	NULL
	5	ashish	NULL	NULL

Result 5 x

Output

Action Output

	#	Time	Action	Message
✓	8	21:28:18	SELECT * FROM class INNER JOIN class_info ON class...	2 row(s) returned
✓	9	21:28:51	SELECT class.name, class_info.address FROM clas...	2 row(s) returned
✓	10	21:29:21	SELECT * FROM class NATURAL JOIN class_info LIMIT...	2 row(s) returned
✓	11	21:29:42	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
✓	12	21:29:59	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANI...	2 row(s) affected Reco
✓	13	21:30:12	SELECT * FROM class LEFT OUTER JOIN class_info O...	4 row(s) returned

Step 7 – Find Unmatched Rows from the Left Table

```
52 • SELECT *
53 FROM class
54 LEFT JOIN class_info
55 ON class.id = class_info.id
56 WHERE class_info.id IS NULL;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name	id	address
▶	4	alex	NULL	NULL
	5	ashish	NULL	NULL

Result 6

Output



Action Output

	#	Time	Action	Message
✓	9	21:28:51	SELECT class.name, class_info.address FROM clas...	2 row(s) returned
✓	10	21:29:21	SELECT * FROM class NATURAL JOIN class_info LIMIT...	2 row(s) returned
✓	11	21:29:42	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
✓	12	21:29:59	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANI...	2 row(s) affected Recd
✓	13	21:30:12	SELECT * FROM class LEFT OUTER JOIN class_info O...	4 row(s) returned
✓	14	21:30:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	2 row(s) returned

Step 8 – RIGHT OUTER JOIN

```
58 • SELECT *
59 FROM class
60 RIGHT OUTER JOIN class_info
61 ON class.id = class_info.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: ☐

	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 7 x

Output

Action Output ▼

	#	Time	Action	Message
✓	10	21:29:21	SELECT * FROM class NATURAL JOIN class_info LIMIT...	2 row(s) returned
✓	11	21:29:42	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
✓	12	21:29:59	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANI...	2 row(s) affected Recon
✓	13	21:30:12	SELECT * FROM class LEFT OUTER JOIN class_info O...	4 row(s) returned
✓	14	21:30:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	2 row(s) returned
✓	15	21:31:46	SELECT * FROM class RIGHT OUTER JOIN class_info ...	5 row(s) returned

Step 9 – Find Unmatched Rows from the Right Table

```
63 • SELECT *
64 FROM class
65 RIGHT JOIN class_info
66 ON class.id = class_info.id
67 WHERE class.id IS NULL;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name	id	address
▶	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 8

Output



Action Output

	#	Time	Action	Message
✓	11	21:29:42	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
✓	12	21:29:59	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANI...	2 row(s) affected Reco
✓	13	21:30:12	SELECT * FROM class LEFT OUTER JOIN class_info O...	4 row(s) returned
✓	14	21:30:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	2 row(s) returned
✓	15	21:31:46	SELECT * FROM class RIGHT OUTER JOIN class_info ...	5 row(s) returned
✓	16	21:32:15	SELECT * FROM class RIGHT JOIN class_info ON class...	3 row(s) returned

Step 10 – Simulate FULL OUTER JOIN

```
69 • SELECT *
70 FROM class
71 LEFT JOIN class_info
72 ON class.id = class_info.id
73 UNION
74 SELECT *
75 FROM class
76 RIGHT JOIN class_info
77 ON class.id = class_info.id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	4	alex	NULL	NULL
	5	ashish	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 9 ×

Output

Action Output ▼

	#	Time	Action	Message
✓	12	21:29:59	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANI...	2 row(s) affected Recc
✓	13	21:30:12	SELECT * FROM class LEFT OUTER JOIN class_info O...	4 row(s) returned
✓	14	21:30:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	2 row(s) returned
✓	15	21:31:46	SELECT * FROM class RIGHT OUTER JOIN class_info ...	5 row(s) returned
✓	16	21:32:15	SELECT * FROM class RIGHT JOIN class_info ON class...	3 row(s) returned
✓	17	21:32:53	SELECT * FROM class LEFT JOIN class_info ON class.i...	7 row(s) returned

Step 11 – FULL OUTER JOIN for Unmatched Rows

```
81  LEFT JOIN class_info
82  ON class.id = class_info.id
83  WHERE class_info.id IS NULL
84  UNION ALL
85  SELECT *
86  FROM class
87  RIGHT JOIN class_info
88  ON class.id = class_info.id
89  WHERE class.id IS NULL;
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content: ☐

	id	name	id	address
▶	4	alex	NULL	NULL
	5	ashish	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 10 ×

Output

Action Output

	#	Time	Action	Message
✓	13	21:30:12	SELECT * FROM class LEFT OUTER JOIN class_info O...	4 row(s) returned
✓	14	21:30:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	2 row(s) returned
✓	15	21:31:46	SELECT * FROM class RIGHT OUTER JOIN class_info ...	5 row(s) returned
✓	16	21:32:15	SELECT * FROM class RIGHT JOIN class_info ON class...	3 row(s) returned
✓	17	21:32:53	SELECT * FROM class LEFT JOIN class_info ON class.i...	7 row(s) returned
✓	18	21:33:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	5 row(s) returned

Step 12 – UNION of Two Tables

```
107  
108 • SELECT *  
109 FROM first_table  
110 UNION  
111 SELECT *  
112 FROM second_table;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name
▶	1	abhi
	2	adam
	3	chester

Result 11



Output



Action Output



	#	Time	Action	Message
✓	18	21:33:40	SELECT * FROM class LEFT JOIN class_info ON class.i...	5 row(s) returned
✓	19	21:34:13	CREATE TABLE first_table (id INT, name VARCHA...	0 row(s) affected
✓	20	21:34:31	CREATE TABLE second_table (id INT, name VAR...	0 row(s) affected
✓	21	21:34:47	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Recc
✓	22	21:35:00	INSERT INTO second_table VALUES (2, 'adam'), (3, 'ch...	2 row(s) affected Recc
✓	23	21:35:23	SELECT * FROM first_table UNION SELECT * FROM se...	3 row(s) returned

Step 13 – UNION on a Single Column

```
114 • SELECT name
115     FROM first_table
116     UNION
117     SELECT name
118     FROM second_table;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	name
▶	abhi
	adam
	chester

Result 12 ×



Output



Action Output



	#	Time	Action	Message
✓	19	21:34:13	CREATE TABLE first_table (id INT, name VARCHA...	0 row(s) affected
✓	20	21:34:31	CREATE TABLE second_table (id INT, name VAR...	0 row(s) affected
✓	21	21:34:47	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Reco
✓	22	21:35:00	INSERT INTO second_table VALUES (2, 'adam'), (3, 'ch...	2 row(s) affected Reco
✓	23	21:35:23	SELECT * FROM first_table UNION SELECT * FROM se...	3 row(s) returned
✓	24	21:35:55	SELECT name FROM first_table UNION SELECT name ...	3 row(s) returned

Step 14 – UNION ALL

```
120 • SELECT *
121 FROM first_table
122 UNION ALL
123 SELECT *
124 FROM second_table;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name
▶	1	abhi
	2	adam
	2	adam
	3	chester

Result 13



Output



Action Output



	#	Time	Action	Message
✓	20	21:34:31	CREATE TABLE second_table (id INT, name VAR...	0 row(s) affected
✓	21	21:34:47	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Rec
✓	22	21:35:00	INSERT INTO second_table VALUES (2, 'adam'), (3, 'ch...	2 row(s) affected Rec
✓	23	21:35:23	SELECT * FROM first_table UNION SELECT * FROM se...	3 row(s) returned
✓	24	21:35:55	SELECT name FROM first_table UNION SELECT name ...	3 row(s) returned
✓	25	21:36:35	SELECT * FROM first_table UNION ALL SELECT * FRO...	4 row(s) returned

Step 15 – COUNT with UNION ALL

```
125
126 • SELECT COUNT(*)
127 FROM
128 (
129     SELECT *
130     FROM first_table
131     UNION ALL
132     SELECT *
133     FROM second_table
134 ) AS A;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
COUNT(*)			
4			

Result 14 x

Output

Action Output

	#	Time	Action	Message
✓	21	21:34:47	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Reco
✓	22	21:35:00	INSERT INTO second_table VALUES (2, 'adam'), (3, 'ch...)	2 row(s) affected Reco
✓	23	21:35:23	SELECT * FROM first_table UNION SELECT * FROM se...	3 row(s) returned
✓	24	21:35:55	SELECT name FROM first_table UNION SELECT name ...	3 row(s) returned
✓	25	21:36:35	SELECT * FROM first_table UNION ALL SELECT * FRO...	4 row(s) returned
✓	26	21:37:09	SELECT COUNT(*) FROM (SELECT * FROM first_t...	1 row(s) returned

Step 16 – INNER JOIN Using Table Aliases

```
160 • SELECT c.id, c.name, ci.address
161 FROM class c
162 INNER JOIN class_info ci
163 ON c.id = ci.id;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name	address
▶	1	abhi	DELHI
	2	adam	MUMBAI

Result 19 ×



Output



Action Output



	#	Time	Action	Message
✓	26	21:37:09	SELECT COUNT(*) FROM (SELECT * FROM first_t...	1 row(s) returned
✓	27	21:37:41	SELECT * FROM first_table INTERSECT SELECT * FRO...	1 row(s) returned
✓	28	21:38:15	SELECT name FROM first_table INTERSECT SELECT n...	1 row(s) returned
✓	29	21:38:33	SELECT * FROM first_table EXCEPT SELECT * FROM s...	1 row(s) returned
✓	30	21:38:50	SELECT name FROM first_table EXCEPT SELECT name...	1 row(s) returned
✓	31	21:39:17	SELECT c.id, c.name, ci.address FROM class c INNER J...	2 row(s) returned

Step 17 – CASE Expression with LEFT JOIN

```
165 • SELECT c.id,  
166         c.name,  
167         CASE  
168             WHEN ci.address IS NULL  
169             THEN 'Address Missing'  
170             ELSE 'Address Available'  
171         END AS Status  
172     FROM class c  
173     LEFT JOIN class_info ci  
174     ON c.id = ci.id;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	id	name	Status
▶	1	abhi	Address Available
	2	adam	Address Available
	4	alex	Address Missing
	5	ashish	Address Missing

Result 20

Output



Action Output

	#	Time	Action	Message
✓	27	21:37:41	SELECT * FROM first_table INTERSECT SELECT * FRO...	1 row(s) returned
✓	28	21:38:15	SELECT name FROM first_table INTERSECT SELECT n...	1 row(s) returned
✓	29	21:38:33	SELECT * FROM first_table EXCEPT SELECT * FROM s...	1 row(s) returned
✓	30	21:38:50	SELECT name FROM first_table EXCEPT SELECT name...	1 row(s) returned
✓	31	21:39:17	SELECT c.id, c.name, ci.address FROM class c INNER J...	2 row(s) returned
✓	32	21:39:51	SELECT c.id, c.name, CASE WHEN ci.a...	4 row(s) returned